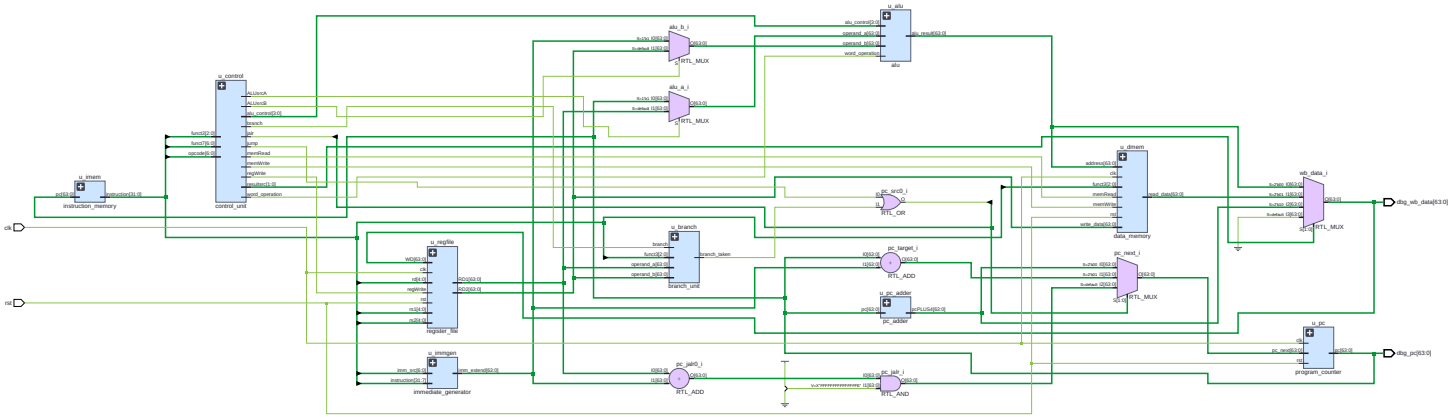
u_alu
alu_control[3:0]
operand_a[63:0]
operand_b[63:0]
word_operation
alu_result[63:0]
alu
alu_b_i
RTL_MUX
alu_a_i
RTL_MUX
u_control
ALUsrcA
ALUsrcB
alu_control[3:0]
branch
jalr
jump
memRead
memWrite
regWrite
resultSrc[1:0]
word_operation
control_unit
u_imem
instruction_memory
u_dmem
data_memory
wb_data_i
RTL_MUX
dbg_wb_data[63:0]
u_branch
branch_taken
branch_unit
pc_src0_i
RTL_OR
pc_target_i
RTL_ADD
u_pc_adder
pc_adder
pc_next_i
RTL_MUX
u_regfile
register_file
u_immgen
immediate_generator
pc_jalr0_i
RTL_ADD
pc_jalr_i
RTL_AND
u_pc
program_counter
dbg_pc[63:0]
clk
rst